

AI-Powered Customer Churn Analytics System

Project Report

Developed for Data Analyst / Power BI Developer Portfolio

Technologies Used:

Python • SQL • Power BI • Machine Learning • Pandas • Scikit-learn

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1. Introduction

Customer churn is a major business challenge for telecom companies. This project uses analytics and machine learning to identify high-risk customers and improve retention strategies.

2. Problem Statement

Telecom companies lose significant revenue due to customer churn. Traditional dashboards only provide historical reports and lack predictive intelligence.

3. Project Objectives

- Analyze customer behavior patterns
- Predict churn using Machine Learning
- Build interactive Power BI dashboards
- Generate business insights and recommendations

4. Technologies Used

- Python
- SQL
- Power BI
- Pandas & NumPy
- Scikit-learn
- GitHub

5. Dataset Overview

The project uses a realistic telecom customer dataset with 5000+ records including customer demographics, payment behavior, internet service details, and churn status.

6. Machine Learning Model

A Random Forest Classification algorithm was used to predict customer churn. The model identifies patterns related to contract type, support tickets, late payments, and monthly charges.

7. Power BI Dashboard

Dashboard pages include:

- Executive Overview
- Customer Behavior Analysis
- AI Prediction Insights
- Revenue Impact Analysis

8. SQL Analysis

SQL queries were used to calculate churn rate, monthly revenue, high-risk customers, and contract-wise churn analysis.

9. Key Business Insights

- Month-to-month customers showed highest churn.
- Customers with repeated late payments had high churn probability.
- High support ticket customers were more likely to leave.
- Long-tenure customers showed better retention.

10. Business Recommendations

- Improve customer support quality
- Launch retention campaigns
- Encourage yearly subscription plans
- Offer loyalty rewards for long-term customers

11. Conclusion

This project demonstrates how Python, SQL, Power BI, and Machine Learning can be combined to solve real-world business problems and support data-driven decisions.

12. Skills Demonstrated

Technical Skills	Business Skills
Python	Customer Retention Analytics
SQL	Business Intelligence
Power BI	KPI Analysis
Machine Learning	Revenue Analytics
DAX	Decision Support

13. Dashboard



